

Steel Plant Delay Analysis & Prediction System

Project Overview

The Steel Plant Delay Analysis & Prediction System is a data analytics dashboard developed to analyze production delays occurring in a steel manufacturing plant. The system collects delay records from multiple departments, stores them in a MySQL database, performs SQL-based analysis, and visualizes insights using an interactive Streamlit dashboard.

The project helps management identify delay patterns, monitor equipment performance, analyze seasonal impacts, and predict possible delay reasons.

Project Objectives

Analyze steel plant production delays.

Visualize delay data through interactive charts.

Perform SQL-based business analysis.

Predict possible delay reasons based on season.

Provide a user authentication system.

Build an easy-to-use management dashboard.

Technology Stack

Frontend

Streamlit

Plotly Express

Backend

Python

Database

MySQL 8.0

Database Connector

mysql-connector-python

Data Processing

Pandas

Authentication

Username & Password Login using Streamlit Session State

IDE

Visual Studio Code

Database Tool

MySQL Workbench

Version

Python 3.13

 Project Folder Structure

Steel-Plant-Delay-Analysis/

```
|
|
| — database/
|   | — 01_create_database.sql
|   | — 02_create_tables.sql
|   | — 03_insert_master_data.sql
|   | — 04_insert_delay_data.sql
|   └─ 05_analysis_queries.sql
|
| — dataset/
|   | — delay_records.csv
|   └─ delay_records.xlsx
|
| — docs/
|   | — ER_Diagram.png
|   └─ Project_Architecture.md
|
| — python/
|   | — assets/
|   |   └─ Rashtriya_Ispat_Nigam.svg
```

```
| |
| |└─ pages/
| |  └─ login.py
| |    └─ dashboard.py
| |
| |└─ app.py
| |  └─ database.py
| |    └─ db_config.py
| |      └─ prediction.py
| |        └─ requirements.txt
| |
| |└─ reports/
| |  └─ Project_Report.pdf
| |
| |└─ screenshots/
| |
| |└─ README.md
```

Database Design

Tables Created

shops

Stores all plant shop information.

Columns

shop_id

shop_name

equipment

Stores equipment details.

Columns

equipment_id

equipment_name

shop_id

conveyors

Stores conveyor details.

Columns

conveyor_id

conveyor_name

agencies

Stores agency names.

Columns

agency_id

agency_name

delay_types

Stores different delay reasons.

Columns

delay_type_id

delay_description

seasons

Stores season information.

Columns

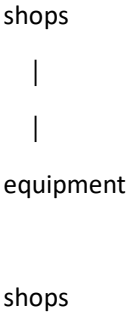
season_id
season_name
delay_records

Main transaction table.

Columns

delay_id
delay_date
shop_id
equipment_id
conveyor_id
agency_id
delay_type_id
season_id
delay_minutes

 Relationships



equipment

conveyors

agencies

delay_types

seasons

|

|

delay_records

⚙️ Project Workflow

User Login

↓

Authentication

↓

Dashboard Access

↓

MySQL Database Connection

↓

Fetch Data

↓

SQL Analysis



Pandas Data Processing



Plotly Visualization



Prediction Module



Logout



Authentication Workflow

User opens application.

Login page is displayed.

User enters Username.

User enters Password.

Credentials are verified.

Session State is created.

Dashboard is opened.

Logout clears Session State.



Dashboard Features

KPI Cards

Displays

Total Delay Records

Total Shops

Total Equipment

Average Delay Minutes

Shop Wise Delay Analysis

Visualization

Interactive Bar Chart

Purpose

Identify shops having maximum delays.

Equipment Wise Delay Analysis

Visualization

Bar Chart

Purpose

Analyze equipment causing delays.

Season Wise Delay Analysis

Visualization

Pie Chart

Purpose

Compare delays across seasons.

Agency Wise Delay Analysis

Visualization

Bar Chart

Purpose

Compare contractor/agency performance.

Conveyor Wise Delay Analysis

Visualization

Bar Chart

Purpose

Identify conveyor-related delays.

Delay Type Analysis

Visualization

Donut Chart

Purpose

Analyze major delay causes.

Monthly Delay Trend

Visualization

Line Chart

Purpose

Observe monthly delay patterns.

Delay Records Table

Displays complete delay information including

Date

Shop

Equipment

Conveyor

Agency

Delay Type

Season

Delay Minutes

 Prediction Module

Simple rule-based prediction.

Summer

Possible Delays

Motor Failure

Hydraulic Failure

Monsoon

Possible Delays

Belt Slip

Material Jam

Winter

Possible Delays

Bearing Failure

 SQL Operations Performed

CREATE DATABASE

CREATE TABLE

INSERT

SELECT

COUNT

AVG

GROUP BY

ORDER BY

JOIN

INNER JOIN

MONTH()

MONTHNAME()

Python Libraries Used

streamlit

mysql-connector-python

pandas

plotly

Pillow

Dashboard Components

Login Screen

Sidebar

Logout Button

Company Logo

KPI Cards

Charts

Prediction Section

Data Table

Future Enhancements

Machine Learning Prediction Model

Real-time Data Integration

Google Authentication

Email Notifications

Export Dashboard Reports

User Role Management

AI-based Delay Prediction

Live Steel Plant Monitoring

✔ Project Outcome

Successfully developed a Steel Plant Delay Analysis & Prediction System capable of:

Managing delay data

Performing SQL analysis

Visualizing business insights

Providing secure login functionality

Predicting delay reasons

Delivering an interactive management dashboard

★ Internship Skills Demonstrated

Database Design

SQL Query Writing

Python Programming

Data Analysis

Data Visualization

Dashboard Development

Streamlit Application Development

Authentication Implementation

Project Structuring

Report Documentation